



BASE24[®]

ERACOM Security Module (ESM) Reference Manual

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ACI (Pacific) Pty. Ltd
1601 Malvern Road
Glen Iris VIC 3146
Australia
Telephone: 61 3 9823-4500
Facsimile: 61 3 9885-0766

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Preface

Reference Material

The following ERACOM documents are referred to :-

*ERACOM SECURITY MODULE
GENERAL SPECIFICATION AND USER MANUAL,
961-500-0112 Rev G 4/87*

*ERACOM SECURITY MODULE
VISA SUPPORT
GENERAL SPECIFICATION AND USER MANUAL,
961-500-0127 Rev B 4/87*

The following BASE24 documents should be referred to :-

*BASE24 TRANSACTION SECURITY
BASE24 LOGICAL NETWORK CONFIGURATION FILE
BASE24 FILES MANUAL*

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Section 1

Eracom Security Module Support Summary

The ERACOM Security Module Support facility provides the software components to support the ERACOM Security Module hardware. This facility provides encryption and key management required in the processing of EFT/POS, ATM and Switch transactions.

The ERACOM Security Module hardware is supported operating in :

- Multiple-Institution mode and
- asynchronous, ASCII mode or,
- ethernet, binary mode.

(Refer *General Specification 961-500-0112 Rev G*).

The purpose of this document is to define the components of the ERACOM Security Module Support facility for BASE24-pos and BASE24-atm which will be referred to as BASE24.

It describes the functions of the components, how they interact, how they are controlled and the interfaces they have that are internal and external to the BASE24 products.

VISA support is an additional option requiring extra, compatible software components and a variation of ERACOM Security Module hardware. (Refer *Supplement to General Specification, VISA Support 961-500-0127 Rev B*).

The Security Module facility allows security device independence in the following BASE24 modules :-

- Device Handler
- Authorisation
- Host Interface
- Switch Interface.

A uniform group of user-callable security utility procedures is provided for performing the following functions :-

- Session Key generation
- Session Key translation
- Message Authentication Code generation
- Message Authentication Code verification
- Personal Identification Number translation
- Personal Identification Number verification
- Card Verification Value generation
- Card Verification Value verification.

The basic structure of the ERACOM Security Module support facility is similar to the Atalla Security Module and VISA Security Module support. The manual procedures required in ESM support differ in some aspects from those that are used to support Atalla Security Modules and VISA Security Modules.

Section 2

Glossary of Terms

Acronyms, Abbreviations, etc.

AFT	Application File Terminal
ASM	Atalla Security Module
AST	Authorisation Selection Table
CAF	Cardholder Account File
CVV	Card Verification Value
EFT	Electronic Funds Transfer
ESM	ERACOM Security Module
HCF	Host Configuration File
ICF	Interchange Configuration File
KVC	Key Verification Code
IDF	Institution Data File
MAC	Message Authentication Code
PIN	Personal Identification Number
POS	Point Of Sale
PSTM	POS Standard Internal Message
PTDF	POS Terminal Data File
SPDH	Standard POS Device Handler
STM	Standard Internal Message
TDF	Terminal Data File
VSM	VISA Security Module

Stored Keys

The following keys are stored in the ERACOM security module :-

MMK	Module Master Key; a key used to encrypt other keys which reside inside the ESM (no BASE24, Atalla, or VSM equivalent)
KM	Domain Master Key; a key used to encrypt short-term session keys which reside in the BASE24 database (BASE24/Atalla equivalent is MFK, Master/Module File Key)
KISn	Interchange Sending Key; the key used to encrypt the PIN/MAC encryption key that is sent TO interchange or institution from BASE24
KIRn	Interchange Receiving Key; the key used to decrypt the PIN/MAC encryption key received FROM interchange or institution (??)(KIS and KIR taken together are equivalent to KEK, or Key Exchange Key; the sender's KIS has the same value as the receiver's KIR)
KBn	Terminal Base Key for terminal n (equivalent to BASE24's Terminal Master Key - used to encrypt the PIN Protection and MAC Processing Session Keys that are downline loaded from BASE24 to the terminal)
PVKn	PIN Verification Key for institution n; associated with Decimalisation Table for institution n, which is also stored in the ESM (PVK is sometimes referred to as KPV)
ZCMK	Zone Control Master Key - the key used to encrypt all working keys under the Visa PVV PIN Verification method.
CVK	Card Verification Keys (A and B). If enabled in BASE24, these keys are used to verify cards during a transaction.

Session Keys

Session keys (KS) are short-term keys which are randomly generated by the ESM and shared by BASE24 and another network entity such as a terminal, a host or a switch.

There are three types of session keys which are used by BASE24 :-

- PPK or PIN Protection Keys,
- MPK or MAC Processing Keys and
- DPK or Data Protection Keys.

A PPK session key shared by BASE24 and an ATM is often called a communication key (KC). A PPK session key shared by BASE24 and an interchange is called a working key, and generally equates to one of two keys :-

- the acquirer working key (AWK) or
- the issuer working key (IWK).

Another general BASE24 term for PPK is PIN Encryption Key (KPE).

Key Variants

As with other security module types, the ESM uses different variants of a key encryption key to protect different types of keys. Specifically, one variant of the domain master key (KM1) is used to protect MPKs and another variant (KM2) is used to protect PPKs.

The variants are calculated using the following constants:

Variant Number	Constant	Used to Protect
1	X' 2424242424242424 '	KMACs, KMACr
2	X' 2828282828282828 '	KPEs, KPEr

Notation

E.X[Y] - The value "Y" encrypted in key "X" is represented in this notation. For example, if a session key (KS) was encrypted under the Domain Master Key (KM) the result would be represented as E.KM[KS].

The form of the encrypted value is usually represented to the user as a 16-hexadecimal-digit representation of eight bytes of encrypted data. In that form it is a "cryptogram" and can be passed as normal ASCII characters.

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Section 3

Design Considerations

The ESM support facility interfaces to ESM devices using asynchronous or ethernet communications. The bisync protocol is not supported.

The ESM support has been based on the ASM and VSM support facilities for BASE24 which use "WRITEREAD" calls to the Tandem GUARDIAN operating system.

The design method obviates impacting on the ASM and VSM implementation, as the current versions of the BASE24 products and Security Utilities adopt less vendor-specific nomenclature and provide greater Security Module transparency to the user than previous versions.

ESM support is facilitated by a separate set of internal security utility procedures pertaining solely to ESM interfacing, which are transparent to these user modules.

To allow a BASE24 network to be configured to use an ERACOM device the following parameters and assigns need to be specified in the appropriate LCONF file.

Parameters:

- SECURE-DEV-TYP
- SECURE-DEV-ACCESS-TYPE
- SECURE-DEV-FAIL-NOTIFY
- SECURE-DEV-REINSTATE-TIME
- SECURE-DEV-TIM-LMT
- SECURE-DEV-TRN-RETRY-CNT
- SECURE-DEV-COMMS-TYPE

Assigns:

- SECURE-DEVn

Refer to section 10 for further information on configuring ERACOM devices.

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Section 4

Module Impact Summary

The Device Handler module transparently call unique ESM utility procedures to access the ESM in order to generate PIN encryption and message authentication keys if the LCONF PARAM SECURE-DEV-TYP indicates that an ESM is to be utilised.

The Host/Switch Interface modules transparently call the PIN translation utility procedures unique to the ASM, VSM or ESM, depending on the LCONF PARAM SECURE-DEV-TYP. PINs from transaction requests are sent to the ESM to be translated using encryption under the Acquirer Working Key (AWK) before they are sent to the host/switch.

The Authorisation module transparently calls unique ESM utility procedures to verify PINs when the LCONF PARAM SECURE-DEV-TYP indicates an ESM is to be utilised. These routines are similar to the ASM and VSM routines but create and send ESM native format verification requests.

The Authorisation module transparently calls ESM- utility procedures to verify cards when the ENCRYPT TYPE in the KEYF file indicates an ESM is to be utilised. These routines are similar to the ASM and VSM routines but create and send ESM native format verification requests.

The AFT system does not require any changes in code to support the ERACOM Security Module. However, the manual procedures for using the screen fields differ from those used for Atalla or VISA security module support.

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Section 5

Transaction Scenarios - PIN Processing

Transaction Originates at BASE24-pos

The transaction request is routed to the Device Handler module by the XPNET node. The PIN block is encrypted in a key which was sent to the terminal encrypted in the terminal base key (KBn) before the terminal was opened.

The Device Handler module determines from the PTDF fields ENCR-PIN.PIN-VAL-TERM and ENCR-PIN.PIN-ENCRYPT-TYP and the LCONF PARAM SECURE-DEV-TYP that the terminal uses ESM key management and PIN security. Therefore it builds a PSTM that in addition to the normal data elements contains:

- a) PIN-FRMT = PIN block format indicator:
 - "0" = clear PIN
 - "1" = ANSI PIN block encrypted
 - "2" = IBM 3624 method (not used)
 - "3" = PIN/pad format encrypted
- b) PIN = E.KS[PIN block] (from terminal request usage)
- c) PIN-KEY = E.KM2[KS] (from PTDF record)

The Router module receives the request and uses the AST to determine whether the request needs to be routed to the Authorisation module or to a Switch Interface process.

If the Router module passes the transaction request to the Authorisation module, then the Authorisation module determines whether the request needs to be authorised at a host, at BASE24, or at a switch.

If the request is to be authorised at a host:

- (a) The Host Interface process receives the request from the Authorisation module. The Host Interface process determines whether the host wants encrypted or clear PINs.
- (b) If the host requires encrypted PINs, the Host Interface process uses an ESM to translate the request PIN from encryption under the terminal's session key into encryption under the session key for the host.

- (c) If the host requires clear PINs, the Host Interface process uses an ESM to translate the PIN into encryption under the intermediate key (KI) and then decrypts the PIN block into the clear, in software. Once in the clear the Host Interface process performs any data formatting required to put the PIN block into the format expected by the host.
- (d) The Host Interface process sends the request to the host. Since the host either approves or denies the request, no further PIN processing is required.

If the request is to be authorised by BASE24-pos, the Authorisation module will use an ESM verification utility to format and send a request to an ESM to verify the encrypted PIN at the ESM.

If the Router module passes the transaction request to a Switch Interface process, the Switch Interface process will use the ESM to translate the PIN block from encryption under the terminal's session key into encryption under the switch's session key. Once the PIN block is translated, the interface process will send the encrypted PIN block to the switch.

Transaction Originates at BASE24-atm

Transaction request is routed to the Device Handler process by the XPNET node. The PIN block is encrypted in a key which was sent to the terminal encrypted in the terminal base key (KBn) before the terminal was opened.

The Device Handler process determines from the TDF fields ENCRYPT-PIN.PIN-VAL-ATM and ENCRYPT-PIN.PIN-ENCRYPT-TYP and the LCONF PARAM SECURE-DEV-TYP that the terminal uses ESM key management and PIN security. Therefore it builds a STM that in addition to the normal data elements contains:

- (a) PIN-FRMT = PIN block format indicator:
 - "0" = clear pin
 - "1" = encrypted ANSI PIN block
 - "2" = encrypted IBM 3624 PIN block (not used by BASE24)
 - "3" = encrypted PIN/PAD PIN block
- (b) PIN = E.KS[PIN block] (from terminal request message)
- (c) PIN-KEY = E.KM2[S] (from TDF record).

Once the Authorisation process receives the request it determines whether the request needs to be authorised at a host, at BASE24, or at a switch.

If the request is to be authorised at a host:

- (a) The Host Interface process receives the request from the Authorisation process. The Host Interface process determines whether the host wants encrypted or clear PINs.
- (b) If the host requires encrypted PINs, the Host Interface process uses an ESM to translate the request PIN from encryption under the terminal's session key into encryption under the session key for the host.
- (c) If the host requires clear PINs, the Host Interface process uses an ESM to translate the PIN into encryption under the intermediate key (KI) and then decrypts the PIN block into the clear, in software. Once in the clear the Host Interface process performs any data formatting required to put the PIN block into the format expected by the host.
- (d) The Host Interface process sends the request to the host. Since the host either approves or denies the request, no further PIN processing is required.

If the request is to be authorised by BASE24-atm, the Authorisation process will use an ESM verification utility to format and send a request to an ESM to verify the encrypted PIN at the ESM.

If the Authorisation process passes the transaction request to a Switch Interface process, the Switch Interface process will use the ESM to translate the PIN block from encryption under the terminal's session key into encryption under the switch's session key. Once the PIN block is translated, the Switch Interface process will send the encrypted PIN block to the switch.

Transaction Originates at Host Computer or Switch

Transaction request arrives at the Interface process.

The Interface process determines whether the host/switch provides clear or encrypted PINs from information in the KEYF file.

If the host/switch sends clear PINs, the Interface process will change the PIN block format if necessary and then send a request to the ESM to encrypt the PIN block under the Issuer Working Key. The IWK is stored encrypted under KM2 in the KEYF file. Once encrypted, the Interface process builds the PSTM/STM with:

PIN-FRMT = PIN block format indicator

PIN = E.IWK[PIN block] - encrypted by the Interface

PIN-KEY = E.KM2[WK] (INBOUND KEYF data)

If the host/switch sends encrypted PINs, the Interface process builds the PSTM/STM and adds:

PIN-FRMT = PIN block format indicator

PIN = E.IWK[PIN block] - PIN block encrypted in the Issuer Working Key
(from the host/switch)
PIN-KEY = E.KM2[IWK] (INBOUND KEYF data)

The request is sent to the Router module (ie BASE24-pos). The Router module will determine whether to send it to Authorisation module or whether to send it to a Switch Interface process for routing to a switch.

If Authorisation module verifies the request, it will take data from the PSTM/STM and its data files and send a PIN verify request to the ESM. The ESM will reply that the PIN was or was not verified.

If the Authorisation module sends the request to a host, the Host Interface process will translate the PIN into the host's session key (AWK ie OUTBOUND of KEYF) via an ESM before forwarding the request to the host. (Refer the previous scenario explanation).

If the request is routed to a Switch Interface process, the interface process will use an ESM to translate the PIN from encryption under the origin host's/switch's session key (IWK ie INBOUND of KEYF) into encryption under the switch's session key (AWK ie OUTBOUND of KEYF).

Section 6

Interface Screen Field Usage

KEYF Impacts

The KEYF file contains one record for each Interface/Interchange process associated with the system. Information on the record contains specific key processing information.

Screen 1 - PIN encryption/translation and MAC keys

BASE24-BASE KEY FILE		____/____/____:____ 01 OF 04	
DPC/FIID: _____		INTERFACE PROCESS: _____	
ENCRYPT TYPE: _ (_____)	BASE24 ENCRYPT TYPE: _ (_____)		
PIN BLOCK FORMAT: _ (_____)	ANSI PAN FORMAT: _ (_____)		
MAC ENCRYPT TYPE: _ (_____)	PIN PAD CHARACTER: _		
MAC DATA TYPE: _ (_____)	NUMBER OF KEYS: _ (_____)		
KEY LENGTH: _ (_____)	FULL MESSAGE MAC: _ (_____)		
INTERMEDIATE KEYS			
CLEAR: _____	CHECK DIGITS: _____		
ENCRYPTED: _____			
EXCHANGE KEYS			
PIN KEY: _____	_____	CHECK DIGITS: _____	
MAC KEY: _____	_____	CHECK DIGITS: _____	
KEKs: _	_____	KVC: _____	
KEKr: _	_____	KVC: _____	
RECORD LAST CHANGED: ____/____/____:____		BY USER: _____, _____ ADD	
***** BASE24 *****			
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:	
	F12-HELP		

Refer *BASE24 BASE FILE MAINTENANCE MANUAL* for a more detailed description.

Fields displayed on screen:

PRIKEY-HCF.DPC-NUM / PRIKEY-ICF.FIID

The DPC number of the host or FIID of the Interchange using this record.

PRIKEY-HCF.HISF-PRO / PRIKEY-ICF.SWI-PRO

The name of the Interface process associated with the DPC or Interchange.

INTERFACE.ENCRYPT-TYP

Specifies how PINs are handled by DPC or Interchange.

INTERFACE.B24-ENCRYPT-TYP

Specifies how PINs are handled internally by BASE24.

INTERFACE.PIN-BLK

Specifies the PIN block format of inbound and outbound PINs.

INTERFACE.ANSI-PAN

Specifies the PAN digits used by the DPC or Interchange in forming ANSI PIN blocks.

INTERFACE.MAC-TYP

Specifies how MACing will be handled between the DPC or Interchange and the BASE24 Interface process.

INTERFACE.PINPAD-CHAR

Specifies the pad character used by the DPC or Interchange in forming PIN/PAD PIN blocks.

INTERFACE.MAC-DATA-TYP

Specifies the character set used for message transmission between DPC or Interchange and the BASE24 Interface process.

INTERFACE.NUM-KEYS

Specifies whether inbound and outbound keys are combined or separate. (Not used by ERACOM-ESM).

INTERFACE.KEY-LGTH

Specifies whether single- or double-length Key Exchange Keys are being used by Interface. (Not used by ERACOM-ESM).

INTERFACE.FULL-MSG-MAC

Specifies whether selected data items or the entire message will be used to create the MAC.

INTERFACE.INTERM.KEY-CLEAR

The clear version of the Intermediate Key.

INTERFACE.INTERM.KEY-CHK-VALUES

Specifies the check digits corresponding to the value in the Encrypted Intermediate Key.

INTERFACE.INTERM.KEY-ENCRYPT

Specifies the security module Encrypted Intermediate key.

INTERFACE.EXCHNG-KEY (-EXTND)

Specifies the security module encrypted form of the key used to exchange PIN keys (single and double length).

INTERFACE.EXCHNG-KEY-CHK-VALUES

Specifies the check digits corresponding to the value in the PIN Key field.

INTERFACE.MAC-EXCHNG-KEY (-EXTND)

Specifies the security module encrypted form of the key used to exchange MAC keys (single and double length).

INTERFACE.MAC-EXCHNG-KEY-CHK-VALUES

Specifies the check digits corresponding to the value in the MAC Key field.

INTERFACE.E^KMV7^KEKS

Specifies the security module encrypted form of the key used to encrypt the PIN and MAC keys sent. (Not used by ERACOM-ESM)

INTERFACE.E^KMV8^KEKR

Specifies the security module encrypted form of the key used to decrypt the PIN and MAC keys received. (Not used by ERACOM-ESM)

Screen 2 PIN/MAC inbound/outbound working keys

BASE24-BASE KEY FILE		____/____/____:____ 02 OF 04	
DPC/FIID: ____		INTERFACE PROCESS: ____	
PIN KEY INFORMATION			
OUTBOUND KEYS:			
PIN KEY1: ____	CHECK DIGITS1: ____	CURRENT INDEX: ____	
PIN KEY2: ____	CHECK DIGITS2: ____	KEY COUNTER: ____	
INBOUND KEYS:			
PIN KEY1: ____	CHECK DIGITS1: ____	CURRENT INDEX: ____	
PIN KEY2: ____	CHECK DIGITS2: ____	KEY COUNTER: ____	
MAC KEY INFORMATION			
OUTBOUND KEYS:			
MAC KEY1: ____	CHECK DIGITS1: ____	CURRENT INDEX: ____	
MAC KEY2: ____	CHECK DIGITS2: ____	KEY COUNTER: ____	
INBOUND KEYS:			
MAC KEY1: ____	CHECK DIGITS1: ____	CURRENT INDEX: ____	
MAC KEY2: ____	CHECK DIGITS2: ____	KEY COUNTER: ____	
RECORD LAST CHANGED: ____/____/____:____ BY USER: _____, _____ ADD			
***** BASE24 *****			
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:	
	F12-HELP		

Refer BASE24 BASE FILE MAINTENANCE MANUAL for a more detailed description.

Fields displayed on screen:

INTERFACE.OUTBOUND.PIN.KEY1

A key used to encrypt (or translate) PINs sent to a DPC or Interchange from BASE24. For ESM under dynamic key management, this field holds the index in the KIS table in the security device.

INTERFACE.OUTBOUND.PIN.KEY-CHK-VALUE1

The check digits corresponding to the value of the PIN field.

INTERFACE.OUTBOUND.PIN.CURR-INDX

Specifies which outbound PIN encryption key is currently being used by the Interface.

INTERFACE.OUTBOUND.PIN.KEY2

A key used to encrypt (or translate) PINs sent to DPC or Interchange from BASE24. For ESM under dynamic key management, this field holds the index in the KIS table in the security device.

INTERFACE.OUTBOUND.PIN.KEY-CHK-VALUE2

The check digits corresponding to the value of the PIN field.

INTERFACE.OUTBOUND.PIN.KEY-CNTR

Specifies the number of times the outbound PIN key has been changed by dynamic key management processing.

INTERFACE.INBOUND.PIN.KEY1

A key used to encrypt PINs sent to BASE24 from the DPC or Interchange. For ESM, under dynamic key management, this field holds the index in the KIR table in the security device.

INTERFACE.INBOUND.PIN.KEY-CHK-VALUE1

The check digits corresponding to the value of the PIN field.

INTERFACE.INBOUND.PIN.CURR-INDX

Specifies which inbound PIN encryption key is currently being used by the Interface.

INTERFACE.INBOUND.PIN.KEY2

A key used to encrypt PINs sent to BASE24 from the DPC or Interchange. For ESM, under dynamic key management, this field holds the index in the KIR table in the security device.

INTERFACE.INBOUND.PIN.KEY-CHK-VALUE2

The check digits corresponding to the value of the PIN field.

INTERFACE.INBOUND.PIN.KEY-CNTR

Specifies the number of times the inbound PIN key has been changed by dynamic key management processing.

INTERFACE.OUTBOUND.MAC.KEY1

A key used to generate MACs sent to DPC or Interchange from BASE24. For ESM, under dynamic key management, this field is not used.

INTERFACE.OUTBOUND.MAC.KEY-CHK-VALUE1

The check digits corresponding to the value of the MAC field.

INTERFACE.OUTBOUND.MAC.CURR-INDX

Specifies which outbound MAC key is currently being used by the Interface.

INTERFACE.OUTBOUND.MAC.KEY2

A key used to generate MACs sent to DPC or Interchange from BASE24. For ESM, under dynamic key management, this field is not used.

INTERFACE.OUTBOUND.MAC.KEY-CHK-VALUE2

The check digits corresponding to the value of the MAC field.

INTERFACE.OUTBOUND.MAC.KEY-CNTR

Specifies the number of times the outbound MAC key has been changed by dynamic key management processing.

INTERFACE.INBOUND.MAC.KEY1

A key used to generate MACs sent to BASE24 from the DPC or Interchange. For ESM, under dynamic key management, this field is not used.

INTERFACE.INBOUND.MAC.KEY-CHK-VALUE1

The check digits corresponding to the value of the MAC field.

INTERFACE.INBOUND.MAC.CURR-INDX

Specifies which inbound MAC key is currently being used by the Interface.

INTERFACE.INBOUND.MAC.KEY2

A key used to generate MACs sent to BASE24 from the DPC or Interchange. For ESM, under dynamic key management, this field is not used.

INTERFACE.INBOUND.MAC.KEY-CHK-VALUE2

The check digits corresponding to the value of the MAC field.

INTERFACE.INBOUND.MAC.KEY-CNTR

Specifies the number of times the inbound MAC key has been changed by dynamic key management processing.

In a static key environment the encrypted PIN keys are entered in the OUTBOUND and INBOUND PIN KEY fields, whereas in a dynamic key environment these fields contain the institution index into the KIS and KIR respectively.

The check digit fields are not currently used by ERACOM.

Screen 3 Dynamic Key Management

BASE24-BASE KEY FILE		____/____/____:____ 03 OF 04	
DPC/FIID: ____		INTERFACE PROCESS: ____	
PIN KEY VARIANT: 0		MAC KEY VARIANT: ____	
LIMITS			
PIN KEY TIMER VALUE: ____		MAC KEY TIMER VALUE: ____	
PIN KEY TIMER INTERVAL: ____ (____)		MAC KEY TIMER INTERVAL: ____ (____)	
PIN KEY TRAN: ____		MAC KEY TRAN: ____	
PIN KEY ERROR: ____		MAC KEY ERROR: ____	
CONSECUTIVE PIN KEY ERROR: ____		CONSECUTIVE MAC KEY ERROR: ____	
		KMAC SYNCHRONIZATION ERROR: ____	
CLEAR OLD KEY TIMER VALUE: ____ (____)			
ORIGINATING ID: ____			
RECEIVING ID: ____			
KEY PROCESSING TYPE: ____ (____)			
NOTARIZATION SUPPORTED: ____ (____)			
RECORD LAST CHANGED: ____/____/____:____		BY USER: _____, _____ ADD	
***** BASE24 *****			
NEW PAGE: _____		FILE DESTINATION: _____	
		NEW LOGICAL NETWORK ID: _____	
F12-HELP			

Refer to *BASE24 BASE FILE MAINTENANCE MANUAL* for a more detailed description.

Fields displayed on screen:

INTERFACE.PIN-KEY-VARIANT

Specifies which key variant to be applied to the PIN Key Exchange Key (KEK) when translating PIN keys.

INTERFACE.MAC-KEY-VARIANT

Specifies which key variant to be applied to the MAC Key Exchange Key (KEK) when translating MAC keys.

INTERFACE.PIN-KEY-TIMER-LMT

Specifies the maximum length of time a PIN key should be used for.

INTERFACE.MAC-KEY-TIMER-LMT

Specifies the maximum length of time a MAC key should be used for.

INTERFACE.PIN-KEY-TRAN-LMT

Specifies the maximum number of transactions that can be performed with the current PIN key, before changing the key.

INTERFACE.MAC-KEY-TRAN-LMT

Specifies the maximum number of transactions that can be performed with the current MAC key, before changing the key.

INTERFACE.PIN-ERR-LMT

Specifies the maximum number of PIN key related translation errors that can occur with the current PIN key before changing the key.

INTERFACE.MAC-ERR-LMT

Specifies the maximum number of MAC verification errors that can occur with the current MAC key before changing the key.

INTERFACE.CONNS-PIN-ERR-LMT

Specifies the maximum number of consecutive PIN key related translation errors that can occur with the current PIN key before changing the key.

INTERFACE.CONNS-MAC-ERR-LMT

Specifies the maximum number of consecutive MAC verification errors that can occur with the current MAC key before changing the key.

INTERFACE.KMAC-SYNC-ERR-LMT

Specifies the maximum number of key synchronisation errors that can occur with the present MAC key before changing the key.

INTERFACE.OLD-KEY-TIMER-LMT

Specifies the maximum length of time (in seconds) that an old key can be used before it is automatically cleared by the Interface process.

INTERFACE.ORG-ID

Specifies the sender of Dynamic Key Exchange messages.

INTERFACE-RCV-ID

Specifies the receiver to DKE messages.

INTERFACE.KEY-PROC-TYP

Specifies the type of Dynamic Key Management processing this Interface can perform.

INTERFACE.NOTARISE-FLG

Specifies whether this Interface supports Key Notarisation.

The Intermediate Keys (Clear and Encrypted) fields are required for ESM support if:

- transaction is outbound from BASE24 to host/switch, and
- the PIN in the PSTM/STM is encrypted, and
- the INTERFACE.ENCRYPT-TYP is set to "0" or "2".

When the Interface process receives a transaction request to be sent to a host/switch expecting clear PINs and that transaction request contains an ESM-secured PIN, the Interface process sends a PIN^TRANSLATE request to the ESM device supplying the Encrypted Intermediate Key as the PPKo (output PIN Protect Key).

The Interface process then decrypts the PIN block in software using the Clear Intermediate Key and sends the transaction request to the host/switch with the PIN in the clear.

The Acquirer Working Key is used to encrypt PINs outbound from BASE24 to a host. This key is stored encrypted under KM2 if ESM specified.

The Issuer Working Key, also stored encrypted under KM2, is used under the following circumstances:

When the Interface process receives a transaction request from an acquirer host whose INTERFACE.ENCRYPT-TYP is set to "1", the Interface process loads the Issuer Working Key field into the PIN-KEY field of the PSTM/STM.

When the Interface process receives a transaction request from an acquirer host whose INTERFACE.ENCRYPT-TYP is set to "2", the Interface process will encrypt the PIN using the ESM via the PIN^TRANSLATE function which accepts a clear PIN in PIN/PAD format and returns it encrypted under E.KM2[PPKo].

E.KM2[PPKo], the Issuer Key is retrieved from KEYF file and the Interface process loads the it into the PIN-KEY field of the PSTM/STM.

An INTERFACE.ENCRYPT-TYP value of "0" or "2" is interpreted to mean that the host/switch sends and/or accepts only clear PINs. Any PIN outbound from BASE24 to the host/switch must be decrypted into the clear. (Refer INTERMEDIATE KEY above).

If the INTERFACE.ENCRYPT-TYP is set to "0" then PINs from the host/switch are left in the clear when loaded into the PSTM/STM. If the INTERFACE.ENCRYPT-TYP is set to "2" then clear PINs are encrypted before loading into the PSTM/STM. (Refer Issuer Working Key above.)

If the host sends and/or accepts only encrypted PINs, then the INTERFACE.ENCRYPT-TYP should be set to "1". Then the Interface process will ensure that all PINs outbound to the host/switch are encrypted under the Acquirer Working Key, and expects that all PINs inbound from the host/switch have been encrypted under the Issuer Working Key.

ICF Impacts

Screen 12 Key Parameters

BASE24-BASE ICF FILE MAINTENANCE		QUA1	FI01	98/09/21	14:34	12 OF 13
INTERCHANGE FIID: SAA		PROCESS: P1A^SAA4				
ICF KEY PARAMETERS						
ATM SETTLE TIME: 23:59 POS SETTLE TIME: 23:59 STAND IN ALLOWED: N (Y/N)						
CUTOVER METHOD: 0 (SEND/RECEIVE 0520) RECON FLAG: 0 (BOTH ATM AND POS)						
ATM DEFAULT ACC: 00 DEFAULT POS DEFAULT ACC: 00 DEFAULT						
SESSION KEY ROLLOVER THRESHOLDS						
KEY EXCHANGE BAD PINS: 336			BAD MACS: 336			
NUM TRANS FOR TOTALS REWRITE: 336			NUM TRANS KEY EXCHANGE: 2336			
SAF TIME UPDATE: N (Y/N)			CALC MAC ON REPEAT: N (Y/N)			
KEY EXCHANGE INFORMATION						INCLUDE
TYPE OF KEY EXCHANGE: 3 (DYNAMIC)						DATA KEY : N (Y/N)
TAPE FILE FOR KEYS:						KVC PROCESSING: Y (Y/N)
TAPE DATE: (YYMMDD)						PROOF ENDPOINT: N (Y/N)
***** BASE24						

NEW PAGE:		FILE DESTINATION:		NEW LOGICAL NETWORK ID:		
		F12-HELP				

Fields displayed on screen:

ATM-SETL-TIM.ATM-SETL-HH / .ATM-SETL-MM

Specifies the hour and minute that SAA should begin ATM reconciliation with its partner.

POS-SETL-TIM.POS-SETL-HH / .POS-SETL-MM

Specifies the hour and minute that SAA should begin POS reconciliation with its partner.

STAND-IN-AUTH

Specifies whether stand-in processing is allowed for the interchange.

CUTOVER-METHOD

Specifies the cutover method for the interchange (0 = send/recv 0520's; 1 = send 0520 only; 2 = receive 0520 only; 3 = no 0520 exchanged).

RECONCILIATION-FLG

Specifies which totals will be sent to the interchange partner at reconciliation, and how many messages will be sent.

DEFAULT-ACC-ATM

Specified the default ATM account when a request is received from an interchange partner.

DEFAULT-ACC-POS

Specifies the default POS account when a request is received from an interchange partner.

TRANS-COUNT-OPTIONS.NUM-BAD-KEYS-PIN

Specifies the number of bad PIN verifications that are to be allowed before initiating a Key Exchange.

TRANS-COUNT-OPTIONS.NUM-BAD-KEYS-MAC

Specifies the number of bad MACs that are to be allowed before initiating a Key Exchange.

TRANS-COUNT-OPTIONS.NUM-TRANS-ICF-REWRITE

Specifies the number of transactions that the SAA Interchange process will omit process before updating the first record (ie. totals record) in the current ILF.

NUM-TRANS-KEY-EXCHANGE

Specifies the number of transactions that the SAA Interchange process will process before initiating a Key Exchange.

SAF-TIM-UPD

Specifies whether the transmit time is updated before sending a SAF transaction to the co-network.

MAC-REPEAT-CALC

Specifies whether the MAC is calculated on the repeat or original transaction when sending a repeat SAF transaction.

KEY-EXCHNG-TYP

Specifies how the SAA Interchange process is to perform its key management (Static, Dynamic or Tape).

INCL-DATA-KEY

Specifies whether the Data Key should be included in the transaction.

TKEY-FILE-NAME

Specifies the name of the Tape File containing the PIN and MAC keys. Only used if the Key Exchange Type field is set to 2 (TAPE KEY FILE).

INCL-KVC

Specifies whether KVC processing is supported by this Interchange.

TAPE-DATE

Specifies the date of the Tape File. This date is compared to that on the Tape File to ensure the correct file is being read. Only used if the Key Exchange Type is set to 2 (TAPE KEY FILE).

PROOF-ENDPOINT

Specifies whether proof of endpoint is supported by this Interchange.

Section 7

Institution Screen Field Usage

KEYA Impacts

The KEYA file contains information required by the Authorisation module for verifying PINs and cards. (Note: PIN verification is controlled via IDF or CPF, card verification is controlled by CPF only).

Screen 1 requests the information required to select the appropriate KEYA record.

Screen 2 DES (IBM 3624) PIN verification

BASE24-BASE KEY AUTH FILE		___/___/___	:___	02 OF 06
GRP: ___ BEGIN DATE (YYYYMM): ***** END DATE (YYYYMM): ***** FIID: ___				
DES (IBM 3624) PIN VERIFICATION				
DECIMALIZATION TABLE: _____	PAN VERIFY OFFSET: _____			
CLEAR KEY: _____	PAN VERIFY LENGTH: _____			
ENCRYPTED KEY: _____	PAN PAD CHARACTER: _____			
CHECK DIGITS: _____				
RECORD LAST CHANGED: ___/___/___:___ BY USER: _____, _____ CHANGE				
***** BASE24 *****				
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:		
	F12-HELP			

Refer BASE24 BASE FILE MAINTENANCE MANUAL for a more detailed description.

Fields displayed on screen:

IBM-DES.DEC-TBL

Specifies the Decimalisation Table used for generating the institution's PINs or PIN offsets. (Not used by ERACOM).

IBM-DES.PAN-VFY-OFST

Specifies the first position of the card number (PAN) to be used in the formation of the validation data used in DES (IBM 3624) PIN verification.

IBM-DES.KEY-CLEAR

Specifies the clear version of the PIN verification key used for performing DES (IBM 3624) PIN verification in software.

IBM-DES.PAN-VFY-LGHT

Specifies the number of positions of the card number (PAN) to be used in the formation of the validation data used in DES (IBM 3624) PIN verification.

IBM-DES.KEY-ENCRYPT

Specifies the encrypted version of the PIN verification key used for performing DES (IBM 3624) PIN verification by a security module. For ESM, the last two digits hold the PVK index (where the PVK is stored in the security device). Valid values are from “01” to “20”.

IBM-DES.PAN-PAD

Specifies the character used to pad validation data used in DES (IBM 3624) PIN verification.

VERIFY.IBM-DES.ENCRYPT-CHK-VALUES

Specifies the check digit corresponding to the value in the Encrypted Key field. (Not used by ERACOM).

The PIN verification key is not stored in the KEYA for ESMs, but rather in the ESM itself. However, the institution index (under which the PVK and Decimalisation Table are stored in the ESM) must be entered into the KEYA. The field IBM-DES.KEY-ENCRYPT is used to hold this index. The index is entered right-justified, with zero-fill on the left. The Pathway code does not edit this field, so the right-justification and zero-fill must be done manually.

For DES(IBM 3624) PIN verification the PIN DATA fields are set up for use in the PIN^VERIFY^IBM^DES user utility.

The field PAN1 which is supplied to the ESM for PIN verification is built by the Authorisation module using IBM-DES.PAN-VFY-OFST, IBM-DES.PAN-VFY-LGHT and IBM-DES.PAN-PAD.

Fields IBM-DES.DEC-TBL and IBM-DES.KEY-CLEAR are not needed for ERACOM support. They may be used in the normal way if software PIN verification is being used in addition to ERACOM PIN verification.

Screen 5 VISA PVV PIN verification

```

BASE24-BASE KEY AUTH FILE      ____/____/____ 05 OF 06

GRP: ____ BEGIN DATE (YYYYMM): ***** END DATE (YYYYMM): ***** FIID: ____

      VISA PVV KEYS
      CLEAR      ENCRYPT      CLEAR      ENCRYPT
1  ____          ____          ____          ____
   CHECK DIGITS: ____          CHECK DIGITS: ____
2  ____          ____          ____          ____
   CHECK DIGITS: ____          CHECK DIGITS: ____
3  ____          ____          ____          ____
   CHECK DIGITS: ____          CHECK DIGITS: ____
4  ____          ____          ____          ____
   CHECK DIGITS: ____          CHECK DIGITS: ____
5  ____          ____          ____          ____
   CHECK DIGITS: ____          CHECK DIGITS: ____
6  ____          ____          ____          ____
   CHECK DIGITS: ____          CHECK DIGITS: ____

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
              F12-HELP

```

Refer BASE24 BASE FILE MAINTENANCE MANUAL for a more detailed description.

Fields displayed on screen:

ABA-VISA.KEY-CLEAR

Specifies (up to) six clear PVK pairs to be used when PIN verification is performed in software.

ABA-VISA.KEY-ENCRYPT

Specifies (up to) six security module encrypted PVK pairs to be used when PIN verification is performed by hardware.

VERIFY.ABA-VISA.KEY-ENCRYPT-CHK-VALUES

Specifies the check digits corresponding to the value in the Encrypt field (above this field on screen).

Note: On screen 5 of the KEYA, the only field used by ERACOM is the first Encrypt Key field. This field will contain the institution index under which the encrypted key pair is stored in the ESM. The index must be entered right-justified, with zero-fill on the left.

Screen 6 Card Verification

BASE24-BASE KEY AUTH FILE			____/____/____:____	06 OF 06
GRP: ____ BEGIN DATE (YYYYMM): ***** END DATE (YYYYMM): ***** FIID: ____				
CARD VERIFICATION				
ENCRYPT TYPE: __ (____)				
	CLEAR KEY	ENCRYPTED KEY	CHECK DIGITS	
1	_____	_____	_____	
2	_____	_____	_____	
KCVV: ____			KVC: _____	
CVV LENGTH: _				
***** BASE24 *****				
NEW PAGE:		FILE DESTINATION:	NEW LOGICAL NETWORK ID:	
		F12-HELP		

Refer BASE24 BASE FILE MAINTENANCE MANUAL for a more detailed description.

Fields displayed on screen:

VERIFY.CV.ENCRYPT-TYP

Specifies the type of keys to be used when performing Card Verification.

VERIFY.CV.KEY-CLEAR

Specifies the clear CVK pair to be used when Card Verification is performed in software.

VERIFY.CV.KEY-ENCRYPT

Specifies the security module encrypted CVK pair to be used when Card Verification is performed by hardware.

VERIFY.CV.KEY-ENCRYPT-CHK-VALUES

The check digits corresponding to the value in each Encrypted Key field (next to this field on screen).

VERIFY.CV.E-KMV4-KCVV

The encrypted form of the Card Verification Key pair as a double length key. (Not used by ESM)

VERIFY.CV.KCVV-KVC

The KVC of Card Verification Key. (Not used by ESM).

VERIFY.CV.CVV-LGTH

The number of CVV digits to check, default is 3. (Not used by ESM).

Note: On screen 6 of the KEYA, the only field used by ERACOM is the Encrypt Key field. This field will contain the institution index under which the encrypted key is stored in the ESM. The index must be entered right-justified, with zero-fill on the left.

CPF Impacts

Screen 2 Card Verification

BASE24-BASE PREFIX: _____	CARD PREFIX _____	PAN LENGTH: _____	____/____/____	____:____	02 OF 22 FIID: _____
PIN INFORMATION					
PIN VERIFICATION KEYA GROUP: _____		PIN CHECK TYPE: ____ (_____)			
MAX PIN TRIES: _____		BAD PIN ACTION: _ (_____)			
CARDHOLDER PIN SELECT: _ (Y/N)		ALGO NUMBER LOC: _ (_____)			
CHECK IF HOST ONLINE PIN: _ (Y/N)		POFST/PVV LOC: _ (_____)			
PIN TRIES RESET OPTION: _ (_____)					
CARD VERIFICATION INFORMATION					
CARD VERIFICATION KEYA GROUP: _____		CV CHECK TYPE: _ (_____)			
CV DATE: _____		CHECK IF HOST ONLINE CV: _ (Y/N)			
TRACK1 CVD OFST: _____		TRACK1 SRVC CODE OFST: _____			
TRACK2 CVD OFST: _____		TRACK2 SRVC CODE OFST: _____			
BAD CV ACTION - TRACK DATA COMPLETE: _ (_____)					
BAD CV ACTION - TRACK DATA UNCERTAIN: _ (_____)					
RECORD LAST CHANGED: ____/____/____:____ BY USER: _____ / _____ CHANGE					
***** BASE24 *****					
NEW PAGE: _____		FILE DESTINATION: _____		NEW LOGICAL NETWORK ID: _____	
F12-HELP					

Refer BASE24 BASE FILE MAINTENANCE MANUAL for a more detailed description.

Fields displayed on screen:

CPFBASE.CV-KEYA-GRP

The value used by the Authorisation module to select the correct KEYA record when performing Card Verification for cards with this prefix.

CPFBASE.CV-CHK-TYP

Specifies when card verification is to be attempted when processing complete and uncertain Track 2 data.

CPFBASE.CV-EFF-DAT

Contains the CVV expiration date. If a card had an expiration date greater than or equal to this date, a CVV value is expected on the card and CVV processing will occur. Otherwise, the card may not have a CVV value and CVV processing will not occur.

CPFBASE.CV-CHK

Specifies whether BASE24 should perform Card Verification during transaction screening prior to sending a transaction to the Host.

CPFBASE.TRK1-CV-OFST

Specifies the position of the Card Verification Digits within the card's Track1 data.

CPFBASE.TRK1-SC-OFST

Specifies the position of the Service Code within the card's Track1 data.

CPFBASE.CV-OFST

Specifies the position of the Card Verification Digits within the card's Track2 data.

CPFBASE.SC-OFST

Specifies the position of the Service Code within the card's Track2 data.

CPFBASE.CV-BAD-DISP[0]

Specifies the action to be taken when complete Track2 data is present and the card's CVD fails to be verified.

CPFBASE.CV-BAD-DISP[1]

Specifies the action to be taken when the condition of the card's Track2 data is uncertain and the card's CVD fails to be verified.

Note: Diebold and Identikey PIN verification are not supported by this release of ERACOM support.

Section 8

POS Terminal Data File (PTDF) Screen Field Usage

The ENCR-PIN.ENCR-KEYS.M-KEY field of the PTDF is used to store an index to the key referred to as the terminal BASE KEY (KB), in ERACOM nomenclature. It matches the index entered at the ERACOM console just prior to entering the terminal base key, and is a number from 1 to 99.

More than 99 terminals may be supported with the same BASE24-pos Device Handler, by using the same base key for multiple terminals.

The index number entered in the ENCR-PIN.ENCR-KEYS.M-KEY field of the PTDF is right-justified and zero-filled by the AFT operator.

To indicate that the terminal uses ESM security for PIN processing the following fields are required:

LCONF PARAM SECURE-DEV-TYP is set to "ESM"
ENCR-PIN.PIN-ENCRYPT-TYP of PTDF is set to "03" or "04"
ENCR-PIN.PIN-VAL-TERM is set to "0".

To indicate that that the terminal uses ESM security for MAC processing the following fields are required:

LCONF PARAM SECURE-DEV-TYP is set to "ESM"
MAC-DATA.MAC-VAL-TERM is set to "0"
MAC-DATA.MAC-ENCRYPT-TYP is set to "03"

The MAC-DATA.ENCR-KEYS.MAS-KEY field of the PTDF contains the terminal index number, as for PIN processing (see above).

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Section 9

ATM Terminal Data File (TDF) Screen Field Usage

The ENCRYPT-PIN.ENCRIPT-KEYS.M-KEY field of the TDF is used to store an index to the key referred to as the terminal BASE KEY (KB), in ERACOM nomenclature. It matches the index entered at the ERACOM console just prior to entering the terminal base key, and is a number from 1 to 99.

More than 99 terminals may be supported with the same BASE24-atm Device Handler process, by using the same base key for multiple terminals.

The index number entered in the ENCRYPT-PIN.ENCRIPT-KEYS.M-KEY field of the TDF is right-justified and zero-filled by the AFT operator.

To indicate that the terminal uses ESM security for PIN processing the following fields are required:

LCONF PARAM SECURE-DEV-TYP is set to "ESM"
ENCRIPT-PIN.PIN-ENCRIPT-TYP of TDF is set to "03"

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Section 10

Security Device Assignment and Configuration

LCONF

The user process uses the PIN^INIT user utility which calls PINI^GET^CONFIG^INFO and retrieves the device names of the primary and secondary ERACOM Security Modules from the LCONF file.

During initialisation, the user process retrieves its symbolic name and the ASSIGNS and PARAMS directed to it by calling the appropriate user utility routine.

If the ASSIGN is not directed to a process, the assignment data is only retrieved if no matching ASSIGN is found.

Parameters

SECURE-DEV-TYP, is used to enable the user process to determine which type of security module is used for PIN and MAC security by the BASE24-pos and BASE24-atm network. This should be set to "ESM".

SECURE-DEV-ACCESS-TYPE, indicates which ERACOM device access scheme has been adopted, namely:

"1" - a round robin access in a multi device environment
"0" - access one of two devices. This is the default.

SECURE-DEV-FAIL-NOTIFY, defines the interval that messages are logged for denied transactions due to security device failures. Default 20 (seconds).

SECURE-DEV-REINSTATE-TIME, defines the time in minutes after which a "DOWNED" security device is to be brought "UP".

SECURE-DEV-TIM-LMT, number of seconds to wait for a response from a security device before a transaction is considered to be timed out. Default 5 (seconds).

SECURE-DEV-TRN-RETRY-CNT, defines the number of retries per transaction (mac generation, pin verification, etc) before rejection. Default is 5.

SECURE-DEV-COMMS-TYPE, defines the type of communications supported by the ESM Module. A value of 0 signifies asynchronous and a value of 1 signifies ethernet. Default is 0.

Assigns

SECURE-DEV1, defines the first security device. ERACOM supports up to 20 devices.

Refer to the , BASE24 Logical Network Configuration File Manual for further details.

Error Handling

The ESM devices are opened and any errors are reported when the routine PINI^CONVERSE^WITH^DEVICES is invoked from the PIN^INIT user utility routine.

Two Device Access

If an ESM key management function fails because the ESM has a hardware error or fails to respond, the routine PINI^CRITICAL^ERROR, invoked from the user process by one of the user utility routines, logs the problem. The primary ESM is then marked as unavailable, and the operation is retried with the secondary ESM. These actions are performed internal to the user utility routines.

The user process continues to send requests exclusively to the secondary ESM until it receives a warmboot command. On receiving a warmboot command, the Device Handler module closes and re-opens the primary and secondary ESMs by calling the PIN^INIT user utility routine, and if successful, the primary ESM is marked as available.

Round Robin Access

ESM functions are tried on a round robin basis when more than two ESM devices are configured in the LCONF and the LCONF PARAM SECURE-DEV-ACCESS-TYPE is set to "1".

If an ESM key management function fails because the ESM has a hardware error or fails to respond, the routine PINI^CRITICAL^ERROR, invoked from the user process by one of the user utility routines, logs the problem. The particular ESM is then marked as unavailable, and the operation is retried with the next available ESM. These actions are performed internal to the user utility routines.

The user process continues to send requests on a round robin basis excluding the downed devices until it receives a warmboot command. On receiving a warmboot command, the user process closes and reopens all ESMs by calling the PIN^INIT user utility routine, and if successful, the downed ESM is marked as available.

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Section 11

POS AS2805 Device Handler Modules

When ESM key management is used, the terminal and the Device Handler module may share two keys:

- a PIN Protection Key (PPK) and
- a MAC Processing Key (MPK).

The ESM and the terminal also share a terminal Base Key (KB).

The Device Handler modules support the use of ESMs by a sequence that generates and sends the desired session keys to the terminal:

- PPK, or PIN protection keys and
- MPK, or MAC processing keys.

To indicate that the terminal utilises ESM support, the following fields are set:

- LCONF PARAM SECURE-DEV-TYP to "ESM"
- PTDF field ENCR-PIN.PIN-ENCRYPT-TYP to "03" or "04", and or
- PTDF field MAC-DATA.MAC-ENCRYPT-TYP to "03".

To process a logon request, the Device Handler module optionally invokes the following options:

- (a) utilisation of the ESM to generate a new encrypted PIN protection key, using ESM function code 41 and/or 44,
- (b) utilisation of the ESM to generate a new encrypted MAC processing key, using ESM function code 42 and/or 45.

The new encrypted keys are sent to the terminal in the logon response.

Messages

The PSTM fields updated by the device handlers to support Atalla, VISA or ERACOM security are:

- PIN
- PIN-FRMT
- PIN-KEY
- PINPAD-CHAR.

When the PIN is encrypted under the terminal session key, and ESM support is being utilised, PIN-FRMT is set to "1" or "3" and PIN-KEY contains the PIN protection key encrypted under KM2.

When the PIN is encrypted under the terminal master key, and ESM support is being utilised, PIN-FRMT is set to "4" or "5" and PIN-KEY contains the terminal master key index.

File I/O and Device I/O

If the ESMs do not exist or the OPENs fail for other reasons, the Device Handler module process may still be required to function.

This is because there may be terminals in the network, driven by this Device Handler module, which do not utilise ESM PIN security. The device handler may continue to process some transactions if this situation occurs.

If all ESMs fail, the Device Handler module logs the condition via the user utilities. Further transactions from this terminal cannot be processed.

The device handler does NOT invoke the process termination utility or `abend` at this stage because the system may have terminals which do not require ESM support.

However, further ESM requests are prevented from generating a stream of log error messages until a subsequent warmboot command is successful in restoring ESM availability.

Injecting Keys into ESMs and POS Terminals

Manual Procedures

The ESM is installed and its Module Master Key and Domain Master Key are entered in accordance with the ESM User Manual.

For each terminal utilising ESM security, the terminal index is entered in the PTDF record for the terminal.

For each terminal utilising ESM security, the terminal Base Key, KB, is entered at the ESM console, along with the terminal index, in accordance with the following manual procedure:

1. The operator enters the proper password at the ESM console to put the ESM into the authorised state.
2. The Main Task Selection Menu is displayed and operator enters F2 to get the Keys Selection Menu.

3. The Keys Selection Menu is displayed and operator enters F8 to get the Enter Terminal Initialisation Key screen.
4. The operator enters the terminal's index number (a value from 1 to 99).
5. The ESM responds by prompting for the first part (Part A) of the Base Key. The operator enters the 8 hex digits which make up Part A of the Base Key.
6. The ESM then prompts for Part B of the Base Key. In order to minimise security risk, Part B may be entered by a different operator, Part A having been blanked out so that the second operator will not see it. The operator enters the 8 hex digits which make up Part B.
7. The operator enters F2 to continue the Base Key entry procedure for the rest of the terminal index numbers, and then repeats steps 4 through 6 until all Base Keys have been entered.
8. The KBs are entered manually into the POS terminals, by the appropriately authorised personnel.

Key Management by Device Handler.

On receipt of a request from a terminal participating in an ESM security implementation, the Device Handler module initiates the following sequence of operations.

A request is sent to the ESM to generate a PPK session key. Two possible forms of encryption may be used:

- (a) Terminal Base Key encryption.
ESM function 41, IT-PPK-GEN, is used at every request for the terminal if the terminal type requires a PPK encrypted under the terminal Base Key.
- (b) Previous PPK encryption.
ESM function 41, IT-PPK-GEN is used only at the logon request if the terminal type requires a PPK encrypted under the previous PPK. For all subsequent requests, ESM function 44, NT-PPK-GEN can be used.

For ESM function 41, IT-PPK-GEN, the request to the ESM contains the index of the terminal Base Key, KB-index.

For ESM function 44, NT-PPK-GEN, the request to the ESM contains a cryptogram of the previous PPK, E.KM2[KSn].

The responses from the ESM contain two cryptograms of the PPK:

ESM function 41, IT-PPK-GEN, returns
E.KBn[KS] and
E.KM2[KS].

ESM function 44, NT-PPK-GEN, returns
E.KSn[KSn+1] and
E.KM2[KSn+1].

The cryptograms are used as follows:

- (a) The Device Handler module places the first of the two cryptograms of the new PPK into the response message.
- (b) In either case, the Device Handler module places the second of the two cryptograms of the new PPK on the database (PTDF) to be used for processing the next message or next sequence of messages which form a cryptographically compatible unit.

A request is sent to the ESM to generate a MPK session key. Two possible modes of encryption may be used:

- (a) Terminal Base Key encryption.
ESM function 42, IT-MPK-GEN, is used at every request from the terminal if the terminal type requires a MPK encrypted under the Terminal Base Key.
- (b) Previous MPK encryption.
ESM function 42, IT-MPK-GEN is used only at the logon request if the terminal type requires a MPK encrypted under the previous MPK. For all subsequent requests, ESM function 45, NT-MPK-GEN can be used.

For ESM function 42, IT-MPK-GEN, the request to the ESM contains the index of the terminal Base Key, KB-index.

For ESM function 45, NT-MPK-GEN, the request to the ESM contains a cryptogram of the previous MPK, E.KM2[KSn].

The responses from the ESM contain two cryptograms of the MPK:

ESM function 42, IT-MPK-GEN, returns
E.KBn[KS] and
E.KM2[KS].

ESM function 45, NT-MPK-GEN, returns
E.KSn[KSn+1] and
E.KM2[KSn+1].

The cryptograms are used as follows:

- (a) The Device Handler module places the first of the two cryptograms of the new MPK into the response message.
- (a) In either case, the Device Handler module places the second of the two cryptograms of the new MPK on the database (PTDF) to be used for processing the next message or next sequence of messages which form a cryptographically compatible unit.

The Device Handler process sends the response message to the terminal.

AS2805 Device Handler PIN Processing

Messages received from the terminal are handled according to the type of PIN processing specified in the database (PTDF).

If the PTDF field ENCR-PIN.PIN-ENCRYPT-TYP is set to "00", the terminal sends a clear PIN. The Device Handler module puts the clear PIN in the PSTM PIN field and sets PIN-FRMT to "0" to indicate a clear PIN.

If the PTDF field ENCR-PIN.PIN-ENCRYPT-TYP is set to "01" or "02", the terminal sends an encrypted PIN but does not use the ESM. The Device Handler module does not use the ESM but decrypts the PIN with software, puts the clear PIN in the PSTM PIN field and sets PIN-FRMT to "0" to indicate a clear PIN.

If PTDF field ENCR-PIN.PIN-ENCR-TYPE is set to "03" or "04" and the LCONF PARAM SECURE-DEV-TYP is set to "ESM", then the Device Handler module puts the encrypted PIN in the PSTM PIN field.

The PSTM field PIN-FRMT is set to either "1" for encrypted ANSI format PIN block or "3" for encrypted PIN/PAD format PIN block, respectively.

The Device Handler module also loads PIN-KEY of the PSTM with the KM2-encrypted PPK from the PTDF field ENCR-KEYS.P-KEY.

If PIN-FORMAT is PIN/PAD the Device Handler module sets PINPAD-CHAR to the proper pad character.

AS2805 Device Handler MAC Processing

Messages received from the terminal are handled according to the type of MAC processing specified in the database (PTDF).

If the PTDF field MAC-DATA.MAC-ENCRYPT-TYP is set to "00" then MAC processing is not performed.

If the PTDF field MAC-DATA.MAC-ENCRYPT-TYP is set to "01" or "02", then the MAC is checked on incoming messages or calculated on outgoing messages by utilising software.

If the PTDF field MAC-DATA.MAC-ENCRYPT-TYP is set to "03", then MAC processing is performed by utilising ESMs.

The Device Handler module uses the PIN^MAC^GENERATE utility to generate a Message Authentication Code (MAC), using the encrypted MAC key, MAC-DATA.ENCR-KEYS.MAC-KEY (from PTDF) and the message text.

On incoming messages, the Device Handler module uses the PIN^MAC^VERIFY utility to verify that the MAC the ESM generates for this incoming message is the same as the MAC appended to the message by the terminal which sent the message. If the MACs are not equal the message is rejected.

On outgoing messages the MAC generated by the ESM for this message is appended to the message text just before sending the message to the terminal.

MAC-DATA.MAC-VAL-TERM is set to "0" indicating that the terminal validates the MAC on the message sent by the Device Handler module.

Section 12

ATM Device Handler Modules

When ESM key management is used, the terminal and the Device Handler process may share a PIN Protection Key (PPK) and a terminal Base Key, KB.

The Device Handler process support the use of ESMs by a sequence that generates and sends the desired session keys to the terminal.

To indicate that the terminal utilises ESM support, the following fields are set:

- LCONF PARAM SECURE-DEV-TYP to "ESM"
- TDF field ENCR-PIN.PIN-ENCRYPT-TYP to "03".

To process a load request, the Device Handler process use the ESM to generate a new encrypted PIN protection key, using ESM function code 41.

The new encrypted key are sent to the terminal in the logon response.

Messages

The STM fields updated by the Device Handler process to support Atalla, VISA or ERACOM security are:

- PIN
- PIN-FRMT
- PIN-KEY
- PINPAD-CHAR.

When the PIN is encrypted under the terminal session key, and ESM support is being utilised, PIN-FRMT is set to "1" or "3" and PIN-KEY contains the PIN protection key encrypted under the KM2.

File I/O and Device I/O

If the ESMs do not exist or the OPENs fail for other reasons, the Device Handler Process may still be required to function.

This is because there may be terminals in the network, driven by this device handler, which do not utilise ESM PIN security. The Device Handler process may continue to process some transactions if this situation occurs.

If all ESMs fail, the Device Handler process logs the condition via the user utilities and marks the terminal whose transaction is being serviced as "down".

The Device Handler process does NOT invoke the process termination utility or abend at this stage because the system may have terminals which do not require ESM support.

However, further ESM requests are prevented from generating a stream of log error messages until a subsequent warmboot command is successful in restoring ESM availability.

Injecting KEYS into ESMs and ATM Devices

Manual Procedures

The ESM is installed and its Module Master Key and Domain Master Key are entered in accordance with the ESM User Manual.

For each terminal utilising ESM security, the terminal index is entered in the TDF record for the terminal.

For each terminal utilising ESM security, the terminal Base Key, KB, is entered at the ESM console, along with the terminal index, in accordance with the following manual procedure:

1. The operator enters the proper password at the ESM console to put the ESM into the authorised state.
2. The Main Task Selection Menu is displayed and operator enters F2 to get the Keys Selection Menu.
3. The Keys Selection Menu is displayed and operator enters F8 to get the Enter Terminal Initialisation Key screen.
4. The operator enters the terminal's index number (a value from 1 to 99).
5. The ESM responds by prompting for the first part (Part A) of the Base Key. The operator enters the 8 hex digits which make up Part A of the Base Key.
6. The ESM then prompts for Part B of the Base Key. In order to minimise security risk, Part B may be entered by a different operator, Part A having been blanked out so that the second operator will not see it. The operator enters the 8 hex digits which make up Part B.
7. The operator enters F2 to continue the Base Key entry procedure for the rest of the terminal index numbers, and then repeats steps 4 through 6 until all Base Keys have been entered.

8. The KBs are entered manually into the POS terminals, by the appropriately authorised personnel.

Key Management by Device Handler

On receipt of a request from a terminal participating in an ESM security implementation, the Device Handler process initiates the following sequence of operations.

A request is sent to the ESM to generate a PPK session key which is based on the Terminal Base Key.

ESM function 41, IT-PPK-GEN, is used at every request for the terminal if the terminal type requires a PPK encrypted under the terminal Base Key.

The responses from the ESM contain two cryptograms of the PPK:

ESM function 41, IT-PPK-GEN, returns E.KBn[KS] and E.KM2[KS].

The cryptograms are used as follows :

- (a) The Device Handler process places the first of the two cryptograms of the new PPK into the response message.
- (b) The Device Handler process places the second of the two cryptograms of the new PPK on the database (TDF) to be used for processing the next message or next sequence of messages which form a cryptographically compatible unit.

Device Handler PIN Processing

Messages received from the terminal are handled according to the type of PIN processing specified in the database (TDF).

If the TDF field ENCR-PIN.PIN-ENCRYPT-TYP is set to "00", the terminal sends a clear PIN. The Device Handler process puts the clear PIN in the STM PIN field and sets PIN-FRMT to "0" to indicate a clear PIN.

If the TDF field ENCR-PIN.PIN-ENCRYPT-TYP is set to "01" or "02", the terminal sends an encrypted PIN but does not use the ESM. The Device Handler process does not use the ESM but decrypts the PIN with software, puts the clear PIN in the STM PIN field and sets PIN-FRMT to "0" to indicate a clear PIN.

If TDF field ENCR-PIN.PIN-ENCRYPT-TYP is set to "03" or "04" and the LCONF PARAM SECURE-DEV-TYP is set to "ESM", then the Device Handler process puts the encrypted PIN in the STM PIN field.

The STM field PIN-FRMT is set to either "1" for encrypted ANSI format PIN block or "3" for encrypted PIN/PAD format PIN block, respectively.

The Device Handler process also loads PIN-KEY of the STM with the KM2-encrypted PPK from the TDF field ENCR-KEYS.P-KEY.

If PIN-FRMT is PIN/PAD the Device Handler process sets PINPAD-CHAR to the proper pad character.

Section 13

Interfaces

The Switch Interface and Host Interface processes are treated together as their use of the ERACOM support is identical. The Interface process may use ESMs to translate any transaction request PINs which arrive at the interface from within the BASE24 security encryption zone.

On receiving a transaction request from the switch/host, the Interface process adds the KM2 encrypted form of the Issuer Working Key (IWK), along with the PIN data received, to the PSTM/STM it builds for each switch/host request.

The Interface process usually requires two PIN encryption keys:

- the Acquirer Working Key (AWK) and
- the Issuer Working Key (IWK).

The changing of keys (PIN and MAC keys) can be performed manually by mutual consent at an agreed time during a period of low transaction volume, or dynamically (static or dynamic key management). This is configured in Screen 3 of the KEYF, however, for the SAA Interchange module, this is configured in Screen 12 of the ICF.

The manual procedures required to change static keys are NOT described in this document.

Under dynamic key management, the user can initiate a Key Exchange by certain events such as the number of bad PIN verifications that are allowed; the number of bad MACs that are allowed; or the number of transactions that can be processed.

Messages

Four of the PSTM/STM fields are used to support PIN security:

- PIN,
- PIN-FRMT,
- PIN-KEY, and
- PINPAD-CHAR.

While the Device Handler Module place data in these fields when BASE24 is the transaction acquirer, the Interface process places data in these fields when BASE24 acts as an intermediate switch or authorising issuer.

File I/O and Device I/O

If the ESMs do not exist or the OPENs failed for other reasons and the process cannot function because the switch requires security device support, then the switch interface process invokes the process termination utility.

The Interface process uses the ESM device to translate PINs from the original terminal PIN protection key into the Acquirer Working Key if BASE24 is the acquirer.

If the transaction is received from the switch/host, the interface process does not use the ESM. The Interface process uses the ESM only to translate PINs into the AWK to be forwarded to the switch.

If all ESMs fail, the switch interface logs the condition via the user utilities and invokes the process termination utility.

Interface Acquirer Processing

When a PSTM/STM is received by the process, it checks the PSTM's/STM's PIN-FRMT field to determine if the PSTM/STM is a transaction request with an encrypted PIN.

The PIN-FRMT field of the PSTM/STM also depicts the PIN block as PIN/PAD or ANSI format.

If the PIN is not encrypted then the interface process will need to execute the following depending on INTERFACE.ENCRYPT-TYP:

- "0" - PINs are in clear, then move request directly if switch/host allow clear PINs
- "1" - PINs are encrypted using ESM, then use intermediate keys to encode PIN block and use ESM to do a PIN translate using INTERFACE.OUTBOUND.PIN-KEY1 or PIN-KEY2
- "2" - PINs are encrypted using software, then encode PIN block and encrypt using INTERFACE.OUTBOUND.PIN-KEY1 or PIN-KEY2

If the PIN is encrypted then the interface process will need to execute the following depending on INTERFACE.ENCRYPT-TYP:

- "0" - PINs are in clear, then Interface process does a PIN translate using intermediate keys and ESM does a software decryption to get PIN in the clear
- "1" - PINs are encrypted using ESM, then use ESM to do a PIN translate using INTERFACE.OUTBOUND.PIN-KEY1 or PIN-KEY2
- "2" - PINs are encrypted using software, then use intermediate keys and ESM to decrypt PIN block and then encrypt using INTERFACE.OUTBOUND.PIN-KEY1 or PIN-KEY2 using software

The Interface process then puts the encrypted/clear PIN into the Switch/Host format message and sends it.

Interface Issuer Processing

The Interface process receives a transaction request from the Switch/Host. The PIN field is encrypted in the IWK.

If the INTERFACE.B24-ENCRYPT-TYP requires PINs to be in the clear then the Interface process will need to execute the following depending on INTERFACE.ENCRYPT-TYP:

- "0" - PINs are in clear move request directly to PSTM/STM
- "1" - PINs are in encrypted using ESM, then Interface process does a PIN translate using intermediate keys and ESM does a software decryption to get PIN in the clear
- "2" - PINs are encrypted using software, then decode PIN block using INTERFACE.INBOUND.PIN-KEY1 or PIN-KEY2

If the INTERFACE.B24-ENCRYPT-TYP requires PINs to be ESM encrypted then the Interface process will need to execute the following depending on INTERFACE.ENCRYPT-TYP:

- "0" - PINs are in clear, then Interface process encodes PIN block using intermediate keys and ESM and does a PIN translation
- "1" - PINs are encrypted using ESM, then place encrypted PIN block directly into PSTM/STM
- "2" - PINs are encrypted using software, then use intermediate keys and ESM to decrypt PIN block and then encrypt using intermediate keys using software

The Interface process builds a PSTM/STM from the information supplied in the Switch request plus the encrypted IWK from the KEYF file.

Once the PSTM/STM is built, the Interface process sends it to the Authorisation process.

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Section 14

Authorisation Module

The Authorisation module uses the internal ESM PIN verification utilities which are called via the PIN^VERIFY^IBM^DES or PIN^VERIFY^ABA^VISA User Security Utilities.

The internal utilities format the PIN verification requests into the ESM native mode command format, issue them to the ESM and process the response.

IBM DES PIN verification is the verification method which is supported and it is described in this document.

Messages

The PSTM/STM message format for the Authorisation module is the same whether the key management security is handled by an ASM, VSM or ESM.

File I/O and Device I/O

By invoking the User Utility PIN^INIT, the Authorisation module determines the ESM device names, opens the device files and reports any errors.

If all ESMs do not exist or neither can be successfully opened, processing does NOT cease. This is because the Authorisation module may be operating in a BASE24 system which does not require ESM support or in which only part of the system requires ESM support.

Processing

PIN verification is performed by the Authorisation module in a similar manner for ASM, VSM or ESM support.

An internal Security Utility routine which is transparent to the user process is provided for interfacing with the ESMs to verify encrypted PINs.

When a POS Standard Transaction Request (PSTM) is received, the Authorisation module checks the PIN-FRMT field of the PSTM to determine if the request has an encrypted PIN.

PIN-FRMT values "1" and "3" indicate the PIN is encrypted.

If the PIN is not encrypted it is verified in software.

The Authorisation module determines the verification type by accessing PIN-VRFY-TYP of the appropriate IDF record. If the PIN is encrypted the Authorisation module uses an ESM and data from the relevant KEYF record to verify the PIN.

The user Security Utility PIN^VERIFY^IBM^DES is invoked to check if the PIN check type is "1".

If PIN check type is "04", then the Authorisation module invokes PIN^VERIFY^ABA^VISA (not described in this document, but formerly implemented elsewhere by ESM^PIN^VISA^PVV).

Other verification types are NOT valid for use with the ESM.

If the PIN is encrypted and the check type is NOT
an IBM DES type
or VISA PVV type,
an error report is logged describing the fault.

Appendix A

ESM Stand-Alone Utilities

SECUCOM

Overview

SECUCOM is a menu driven utility which performs various key manipulation functions.

SECUCOM was designed to facilitate the manipulation of security keys. SECUCOM accesses an ERACOM Security Module (ESM) to perform a number of these functions directly or through a Blocker process.

The major functions that SECUCOM supports are:

- DES Encryption and Decryption without using an ESM
- Interchange key generation and translation using an ESM
- Visa PIN translation using an ESM
- Generation of a key file from an interchange tape.

To run SECUCOM:

```
[ RUN ] SECUCOM [ / <run-params> ... / ] [ <eracom-device-name> ]
```

```
[ RUN ] SECUCOM [ / <run-params> ... / ] [ <blocker-process-name> ]
```

Where <eracom-device-name> is the sysgen name of an ERACOM device. If this field is absent then SECUCOM will still run but its functions will be limited to those functions that do not require access to an ESM.

The name of the ESM can be changed from within SECUCOM.

Example:

```
SECUCOM / pri 155, cpu 0 / $ESM01
```

or simply:

```
SECUCOM $TRM2.#A
```

or if the Blocker process is configured.

Where <blocker-process-name> is the PPD name of a blocker process. If this field is absent then SECUCOM will still run but its functions will be limited to those functions that do not require access to an ESM.

The name of the Blocker process can be changed from within SECUCOM.

Example:

SECUCOM / pri 155, cpu 0 / \$BKR1

or simply:

SECUCOM \$BKR1

Features

Features of SECUCOM:

- Easy to follow menu driven screens.
- Ability to change ESM device name without stopping SECUCOM.
- Storage variables within SECUCOM where keys can be stored for later use or displayed,. thus eliminating the need to write keys on paper and re-enter them.
- Generation and displaying of verification codes for all keys.
- Special security "HIDE" option that hides keys when entering and prevents keys from being displayed.
- Low level interface with ESM. Allows commands to be directly interchanged with the ESM. SECUCOM will generate the BCC and decode any errors from the ESM.
- Ability to change and store within variables other information used in key manipulation such as Institution Index, PVK Index, Key type (PIN or MAC) etc.

Main Menu

Once SECUCOM has been started the following screen will be displayed. From this Main Menu various other menus or functions can be initiated.

Main Menu	
1	DES Encrypt / Decrypt / XOR Menu
2	Interchange Functions Menu
3	Interchange Tape Functions Menu
4	VISA Functions Menu
5	Talk directly to ESM
6	Change / Display Variables (or ESM device) Menu
7	Calculate a PIN from a PAN
P	Previous menu
E	Exit
Enter Option or press Ctrl-y for previous menu:	

From the Main Menu other menus are invoked with the touch of a single key. The Main Menu options are:

Option 1

This option will display the "DES Encrypt / Decrypt / XOR Menu".

Option 2

This option will display the "Interchange Functions Menu".

Option 3

This option will display the "Interchange Tape Functions Menu".

Option 4

This option will display the "VISA Functions Menu".

Option 5

This option will invoke the "Talk directly to ESM" function.

Option 6

This option will display the "Change / Display Variables (or ESM device) Menu".

Option 7

This option will invoke the "Calculate a PIN from a PAN" function.

Option P

This option will reprint the "Main Menu".

Option E

This option will exit the SECUCOM program.

Encryption / Decryption / XOR Menu

```
-----
                        DES  Encrypt / Decrypt / XOR Menu
1                        Encrypt Using DES algorithm
2                        Decrypt Using DES algorithm
3                        XOR one key with another key
P                        Previous menu
E                        Exit
Enter Option or press Ctrl-y for previous menu:
-----
```

From this menu the user can encode a key with another key, decode a key with another key or XOR a key with another key. All these functions are performed within software and therefore do not use or require an ESM.

Option 1

After selecting this option, the user will be prompted for a key to encrypt. The user may enter the 16 character hex key or enter a variable name. (Note the variable name should be previously loaded if this option is selected.)

After the user has entered the key to encrypt or a variable name containing that key the user will be prompted for an encryption key. Again the user can enter the key or the name of a previously loaded variable that contains the key.

Once both keys have been entered then SECUCOM will encrypt the first key with the second key using the DES encryption method in software. (Note an ESM is not used). The encrypted key will be displayed and the user will be prompted to store the encrypted key. If the user does not want to store the encrypted key then by pressing the return key the user will be returned to the menu.

If the user decides to store the encrypted key then the variable name is entered.. There are 8 user variables (named U1 through to U8) any of which may be used. There are also some system variables which SECUCOM uses in other functions, these too can be used.

Option 2

After selecting this option the user will be prompted for a key to decrypt. The user may enter the 16 character hex key or enter a variable name. (Note the variable name should be previously loaded if this option is selected.)

After the user has entered the key to decrypt or a variable name containing that key the user will be prompted for a decryption key. Again the user can enter the key or the name of a previously loaded variable that contains the key.

Once both keys have been entered then SECUCOM will decrypt the first key with the second key using the DES decryption method in software (Note an ESM is not used). The decrypted key will be displayed and the user will be prompted to store the decrypted key. If the user does not want to store the decrypted key then by pressing the return key the user will be returned to the menu.

If the user decides to store the decrypted key then the variable name is entered. There are 8 user variables (named U1 through to U8) any of which may be used. There are also some system variables which SECUCOM uses in other functions.

Option 3

After selecting this option the user will be prompted for a key to XOR. The user may enter the 16 character hex key or enter a variable name. If the user enters a variable name then that variable should have been previously loaded.

After the user has entered the key to XOR or a variable name containing that key the user will be prompted for a key to XOR it with. Again the user can enter the key or the name of a previously loaded variable that contains the key.

Once both keys have been entered then SECUCOM will XOR the first key with the second. The XORed key will be displayed and the user will be prompted to store the XORed key. If the user does not want to store the XORed key then by pressing the return key the user will be returned to the menu.

If the user decides to store the XORed key then the variable name is entered. There are 8 user variables named (U1 through to U8) any of which may be used. There are also some system variables which SECUCOM uses in other functions, these too can be used.

Option P

This option will return the user to the "Main Menu".

Option E

This option will exit the SECUCOM program.

Note: The results of these functions will be displayed even if the Display Mode is set to HIDE.

Interchange Functions Menu

Interchange Functions Menu		
1	eKISn[KS]	Generate KS (session key) encrypted under KIS (sending interchange key). This key is given to the interchange partner.
	eKM[KS]	Encrypt KS under KM (master key). This key is used by BASE24 in the KEYF file.
2	eKM[KS]	Decrypts received KS under KIR and then encrypts it under KM. This key is used by BASE24 in the KEYF file.
P		Previous menu
E		Exit
Enter Option or press Ctrl-y for previous menu:		

From this menu the user can generate interchange session keys to be sent to an interchange partner and loaded in the BASE24 KEYF file, or translate a key received from an interchange partner so it can be loaded in the BASE24 KEYF file. Both these functions require an ESM.

Option 1

This option will use the ESM to generate a random session key (KS) and encrypt it under the sending interchange key (KISn). This key will be displayed and stored in variable KS. A second key is also returned which is the random session key encrypted under KM. It is stored in variable EKS.

Option 2

This option will prompt the user for a session key (KS) which it will decrypt under the receive interchange key (KIRn) and encrypt under the KM for storage within BASE24 KEYF file. This key will be stored in variable EKS.

Option P

This option will return the user to the "Main Menu".

Option E

This option will exit the SECUCOM program.

Note: The index used for KIS and KIR is in variable "Institution Index" and can be displayed or set from the "Display or set Variables Menu". The default value is 1.

The keys returned can be either MAC or PIN keys. This is determined by the variable "Key Type" and can be displayed or set from the "Display or set Variables Menu". The default value is PIN.

Tape Functions Menu

Tape Functions Menu	
1	Read interchange tape and generate disc file.
2	Generate interchange tape and disc file.
P	Previous menu.
E	Exit.
Enter Option or press Ctrl-y for previous menu:	

From this menu the user can generate the KEY file (used by SAA Interchange module) from a tape received from an interchange partner or generate the file and tape to send to an interchange partner.

Option 1

Use this option to generate a disc file containing the 256 keys used by the SAA Interchange module. When the user select's this option the user will be prompted for the exchange key. This is the same key that is on the KEYF file, it is used to encrypt the keys before writing them to disc.

The user will then be prompted for the name of the tape drive and the disc file. SECUCOM will read the first record off the tape which contains the FIID and date. The FIID and date will be displayed. The date is to be entered into the KEYF file. The user will be asked for a new FIID.

At this stage the user can press return to use the FIID on tape or enter a new FIID. SECUCOM will then read the tape and create the disc file.

Option 2

This option will generate a tape and disc file containing random keys. The tape is to be sent to the user interchange partner. The disc file is to be used by BASE24.

Option P

This option will return the user to the "Main Menu".

Option E

This option will exit the SECUCOM program.

Note: The KIS and KIR must be loaded in the ESM prior to initiating this function. These keys are obtained from the interchange partner.

The index used for the KIS and KIR are displayed at the start of option 1. The default value is 1.

VISA Functions Menu

Visa Functions Menu		
1	eAWK[PIN]	Translate a PIN from the local key (PPK) to the Visa Acquirer Key (AWK).
2	ePPK[PIN]	Translates a PIN from Visa Issuer Key (IWK) to the local Key (PPK).
3	CVV	Generates a Card Verification Value from the PAN, EXPIRY DATE and SERVICE CODE (taken from card).
P		Previous menu
E		Exit
Enter Option or press Ctrl-y for previous menu:		

This menu allows the user to perform two VISA PIN translations and allows the generation of a CVV from a card's Track2 data.

Option 1

This option will translate from a local key (PPK) to VISA acquirer key (AWK). The user will first be prompted for the PPK then for the IWK. SECUCOM will then translate the keys and display the result which is stored in variable AWK.

Option 2

This option will translate from a VISA issuer key (IWK) to the local key. The user will first be prompted for the IWK then for the eKM1[PPK]. SECUCOM will then translate the keys and display the result which is stored in variable PPK.

Option 3

This option will take the card's PAN, EXPIRY DATE and SERVICE CODE and return a three digit Card Verification Value (CVV) that should be included in VISA Track2 data. The user will first be prompted for the Card's PAN (16 digits only), then the Card's Expiry date (YYMM), and finally the Service Code (3 digits). The ESM will be accessed to calculate the CVV and the screen will display it regardless as to the value of the Display Mode variable.

Option P

This option will return the user to the "Main Menu".

Option E

This option will exit the SECUCOM program.

Talk Directly to ESM Function

This function is supplied as a low level interface with the ESM. It allows command to be directly passed to the ESM.

At the "<" prompt enter an ESM command. To return to the "Main Menu" enter ctrl-y at the prompt.

The BCC is not entered as SECUCOM will calculate this for the user. SECUCOM will decode any errors that the ESM returns.

Example: The user has decided to generate a random initial interchange key using institution index 3 in the ESM. This is function 51. The user would enter "5103" at the "<" prompt. Incidentally this is option 1 from the "Interchange Functions Menu".

Display or set Variables Menu

	Change System Variables Menu
1	Display all system variables
2	Change / Copy a saved key
3	Change Display mode (hide or show keys)
4	Change Institution index
5	Change PIN index
6	Change key type (PIN or MAC)
7	Change ESM device
P	Previous menu
E	Exit
Enter Option or press Ctrl-y for previous menu:	

Option 1

This option will cause SECUCOM to display all its variables.

Option 2

This option is used to enter a key, or PAN, into a variable. When prompted, the user may enter the 16 characters of the key or PAN, or a variable name. By entering a variable name the contents are copied from one variable to another.

The user will then be prompted for a variable to store the key, or PAN, in.

Note: The user can only store a PAN in the User Variables named U1 through U8. Keys can be stored in any User Variable.

Option 3

This option is used to change the display mode. Enter "H" to use mode HIDE, "S" to use mode SHOW or "A" to use mode HIDE ALL. If the user select's the HIDE option then keys will not be displayed as they are entered, and will be displayed as "**** * **** * **** * **** * " by SECUCOM. The only exception to this is when the results of the encryption of keys is displayed, unless the display mode has been set to HIDE ALL. PINs and CVVs are always shown in the clear regardless of display mode. The default mode is SHOW.

Option 4

This is used to change the "Institution Index" which is used by some SECUCOM functions. It is the index of the KIS and KIR. The default value is 1.

Option 5

This is used to change the "PIN Index" which may be used in future versions. The default value is 1.

Option 6

This is used to change the "Key Type". Enter "P" for PIN or "M" for MAC. PIN keys and MAC keys are treated differently by the ESM. The default value is PIN.

Option 7

This option is used to define or change the current ESM device. Enter the SYSGEN name of an ESM. There is no default for this variable, but it can be specified at run time.

Option P

This option will return the user to the "Main Menu".

Option E

This option will exit the SECUCOM program.

Calculate a PIN from a PAN

This function is supplied to allow a Personal Identification Number (PIN) to be calculated from a 16 digit Primary Account Number (PAN).

After selecting this option the user will be prompted for a PAN. The user may enter the full 16 digits of the PAN, or the name of a previously loaded user variable.

After the user has entered the PAN, or a user variable containing the PAN, the user will be prompted for the PIN Verification Key (PVK). Again the user can enter all 16 digits of the hex key or the name of a previously loaded user variable.

At the prompt for the Decimalisation Table the user can simply press <enter> to accept the default table of "0123456789012345", or the user can enter all 16 digits manually.

Once SECUCOM has valid values for the PAN, PVK and Decimalisation Table, the PIN is calculated.

The screen will show the value of the PIN both before and after decimalisation, together with the prompt to return to the Main Menu.

Note: The PIN is always displayed in the clear regardless as to the setting of the Display Mode variable. Currently, SECUCOM only supports 16 digit PANs.

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Appendix B

Configuring ESM Security Devices

There are two types of communication configurations used to connect ESM devices to a Tandem system asynchronous and ethernet. The type used depends on the interface option installed in the ESM.

Asynchronous Connection Configuration

The asynchronous configuration requires a dedicated Tandem asynchronous port to be configured and directly connected to the ESM via a cable supplied by Eracom.

This is an example of the SYSGEN parameters required to configure an asynchronous port on a multi function controller to be used be an ESM. In this example the port name of the ESM is \$T3.#A. This name is entered in the LCONF file.

```
#####  
MULTIFUNCTION_CONTROLLERS:  
#####  
  
MFC1      3681      0,1      %140;  
  
#####  
CONTROLLERS:  
#####  
  
COMMB      7700      MFC1      %020;  
  
#####  
PERIPHERALS:  
#####  
  
$T3      COMMB.4, COMMB.5      ASYNCPROC 6401 EIA, NOECHO,      ! Eracom II  
                                BAUD9600, LINETERM %76,      NAME #A;
```

Ethernet Connection Configuration

To use the ethernet option, an existing Local Area Network (LAN) must have been configured on the Tandem system.

The ethernet configuration requires configuring addition ports on the LAN configuration.

This is an example of a SCF input file used to configure an ESM. This example assumes a LAN called \$LANA has already been configured and started.

```
COMMENT This file configures LAN ports to be used by the Eracom
COMMENT Security Module.
Comment
allow 10 errors
assume line $LANA
ADD PORT #REC01, TYPE LLC1, ADDRESS %H0004, DELIVERYMODE SERIAL,&
    REQUESTSIZE 31000, DATAFORWARDTIME 0.060, DATAFORWARDCOUNT 1
ADD PORT #REC02, TYPE LLC1, ADDRESS %H0008, DELIVERYMODE SERIAL,&
    REQUESTSIZE 31000, DATAFORWARDTIME 0.060, DATAFORWARDCOUNT 1
ADD PORT #REC03, TYPE LLC1, ADDRESS %H000C, DELIVERYMODE SERIAL,&
    REQUESTSIZE 31000, DATAFORWARDTIME 0.060, DATAFORWARDCOUNT 1
ADD PORT #REC04, TYPE LLC1, ADDRESS %H0010, DELIVERYMODE SERIAL,&
    REQUESTSIZE 31000, DATAFORWARDTIME 0.060, DATAFORWARDCOUNT 1
ADD PORT #REC05, TYPE LLC1, ADDRESS %H0014, DELIVERYMODE SERIAL,&
    REQUESTSIZE 31000, DATAFORWARDTIME 0.060, DATAFORWARDCOUNT 1
ADD PORT #REC06, TYPE LLC1, ADDRESS %H0018, DELIVERYMODE SERIAL,&
    REQUESTSIZE 31000, DATAFORWARDTIME 0.060, DATAFORWARDCOUNT 1
START PORT *
```

Appendix C

Ethernet Communications

Introduction

Eracom provide the facility of using Ethernet communications with an ESM device. This Appendix outlines the installation guidelines for consideration.

Tandem Hardware / Software

The Tandem hardware and software requirements with installation guidelines are outlined in Appendix D of the Eracom Security Control Module (SCM) Reference Manual.

Eracom Configuration

The installation guidelines for the ESM are outlined in Appendix D of the Eracom Security Control Module (SCM) Reference Manual.

a) Host Configuration

Access the Host Port Configuration screen from the Main Task Selection menu.

Ethernet Address

The Eracom must be configured with its own Ethernet address. At the time of manufacture a default local address will have been included and this may be used.

Programmatic Interface

The programmatic interface is IEEE 802.2 compliant (Logical Link Control Type 1) and this function must be enabled.

b) Security Keys

Domain Master Key (KM)

This is a double length key which is made up from the XOR of two components.

Transaction Keys

All other security keys (PVKs, DTs, KISs, KIRs, KBs, etc.) are single length transaction keys.

Blocker Configuration

The Blocker process provides the application interface to the Ethernet / LLC1 network.

The installation guideline and details on configuration for the Blocker process are outlined in Appendix E of the Eracom Security Control Module (SCM) Reference Manual.

In addition, note the following with regard to LCONF parameters.

BLKR-SDU-BLOCKING-COUNT

This parameter must be set to “1” so that only one request sent to the security module on an individual write.

BLKR-SCM-TYP

This is a new parameter which indicates the type of functionality of the security module, where the valid values are “MARKII” and “APCA” (default). The support of this security module requires that the parameter is set to “MARKII”.

BASE24 Configuration

The configuration guidelines for BASE24 are outline in Section 10 of this manual. In addition, note the following with regard to LCONF assignments and parameters.

SECURE-DEVx

These assignments should be set to the PPD name of the Blocker processes.

SECURE-DEV-COMMS-TYPE

This is a new parameter which indicates the type of communications to the security module, where the valid values are “0” for asynchronous communications (default) and “1” for ethernet communications. The support of this security module requires that the parameter is set to “1”.

SECUCOM

Secucom is a stand-alone utility which provides means for accessing the security module without having to interface with BASE24 or use devices to generate transactions. To use this utility simply run BA50ESM.SECUCOM with the PPD name of the Blocker process as a run-time parameter. Refer to Section A of this manual for more information.